5 **(a)** **(i)** State what is meant by the terms *electric field* and *electric field strength*.

electric field …….....

……..……

electric field strength .....

……..... [2]

For Examiner's Use

(ii) Determine the electric field strength at a distance of 25 cm from a point charge of 5.2×10^{-7} C. Give a unit for electric field strength with your answer.

electric field strength = ... unit ..... [3]

(b) Fig. 5.1 shows three charges of value 1.0 μC at X, −1.0 μC at Y and 1.0 μC at Z. These charges are at the corners of an equilateral triangle.

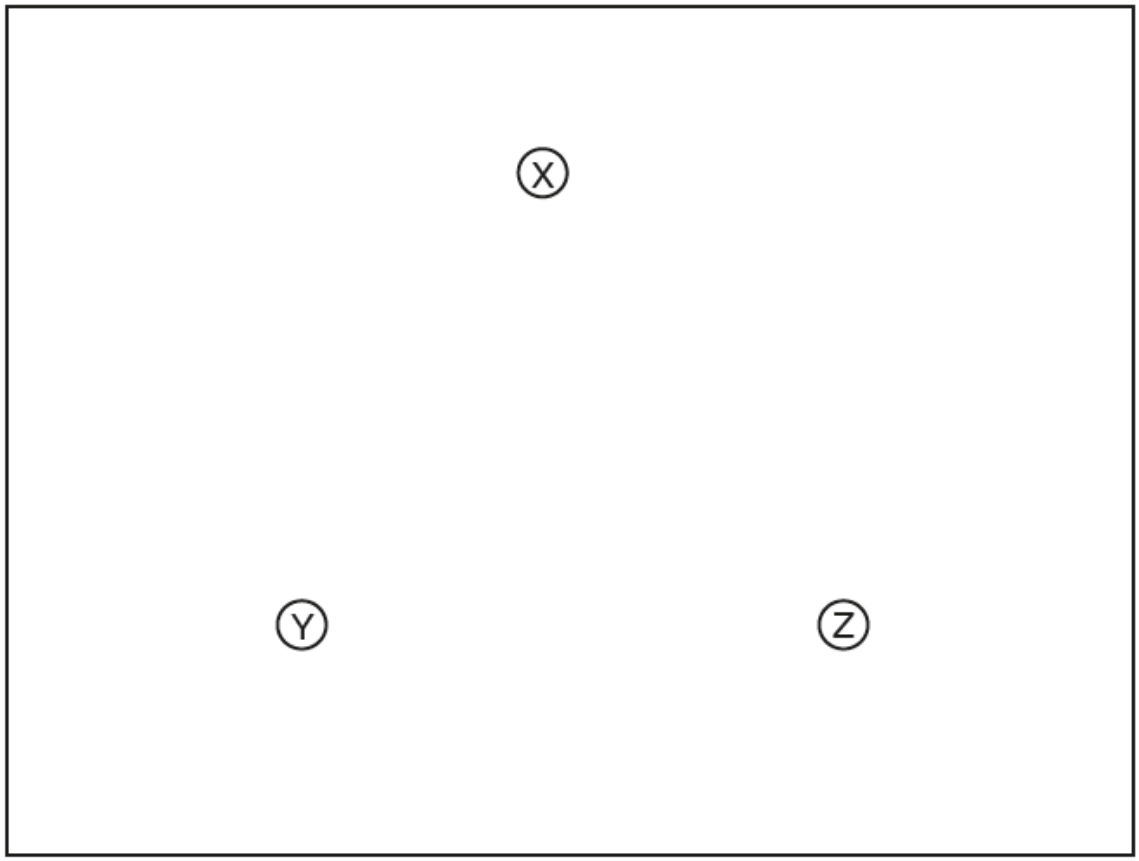

Fig. 5.1

Without making any calculations, draw on Fig. 5.1, the electric field, indicating its main characteristics, within the given rectangle area. [3]